



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 30 summarizes the current fairness state for upload 30. Accuracy changed to +0.782 and fairness score changed to +0.933. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.873.

ACCURACY

0.782

▲ +0.782

FAIRNESS SCORE

0.933

▲ +0.933

COMBINED SCORE

0.873

ThresholdOptimizer_mitigated_threshold

DISPARATE IMPACT

0.732

Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	compas_dt_baseline.pkl	comparison

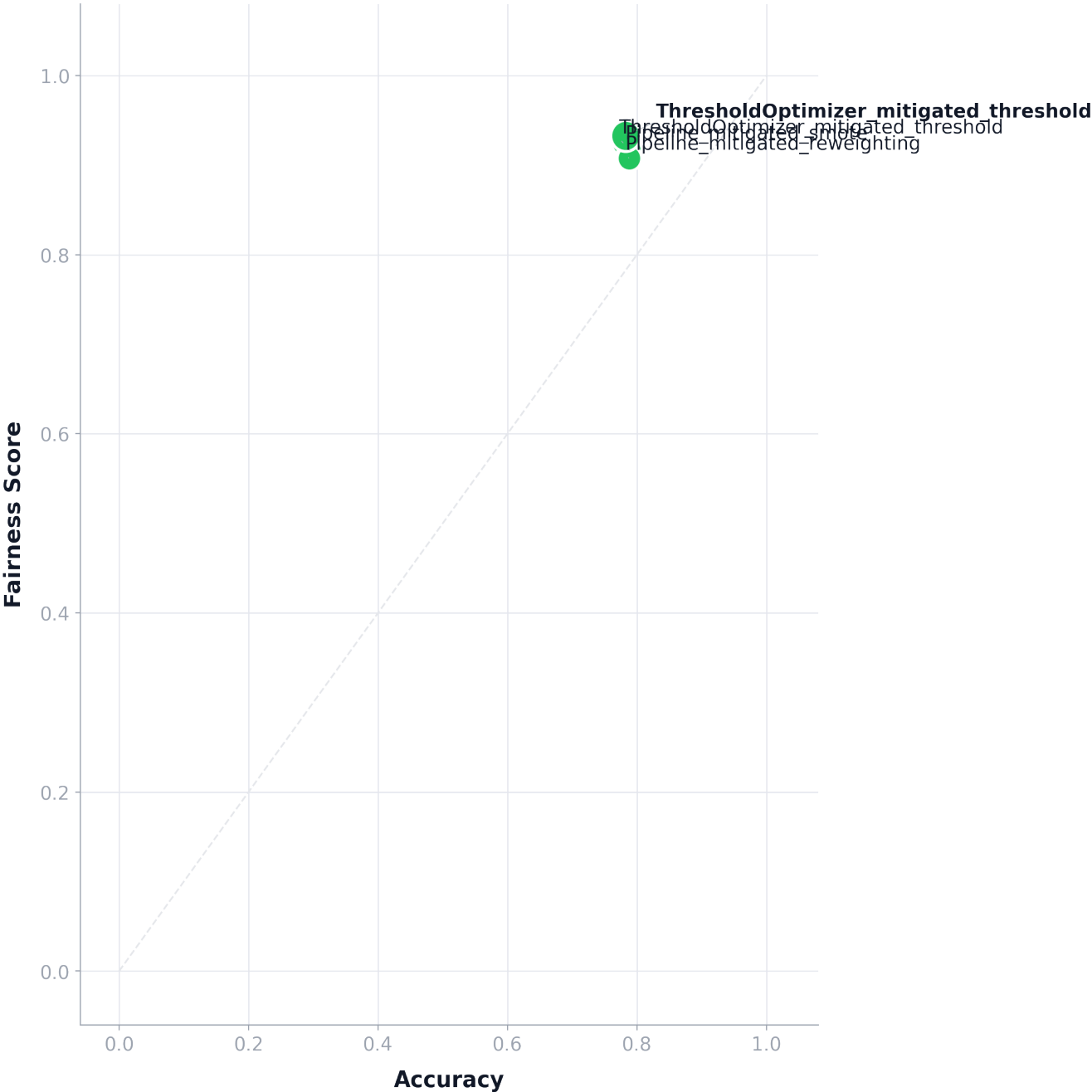
Fairness Metrics

Metric	Value	Interpretation
Demographic Parity Difference (DPD)	0.109	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.025	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.732	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.782	0.933	0.109	0.025	0.732	0.873
★ ThresholdOptimizer_mitigated_threshold	0.779	0.925	0.106	0.044	0.739	0.866
Pipeline_mitigated_smote	0.789	0.916	0.139	0.028	0.672	0.866
Pipeline_mitigated_reweighting	0.788	0.908	0.144	0.040	0.654	0.860

Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.